



Semester 1 Examinations 2020 / 2021

Exam Code(s) 1MF1, 1SD1, 1SD3, 1DAV1
Exam(s) Master's in Software Design & Development
Higher Diploma in Applied Science (Software Design & Development)
Higher Diploma in Applied Science (Software Design & Development – Industry stream)
Higher Diploma in Data Analytics and Visualisation

Module Code(s) CT870
Module(s) Internet Programming

Paper No. 1

External Examiner(s) Prof. Pier Luca Lanzi
Internal Examiner(s) Prof. Michael Madden
*Dr. Enda Barrett

Instructions: Answer any 3 questions. All questions carry equal marks.

Duration 2hrs
No. of Pages 4 (Including cover page)
Discipline Computer Science

Requirements None

Disclaimer

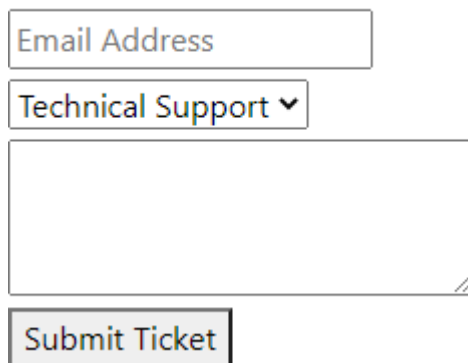
This is an “open book” style exam.

In submitting this work I confirm that it is entirely my own. I acknowledge that I may be invited to online interview if there is any concern in relation to the integrity of my exam, and I am aware that any breach will be subject to the University's Procedures for dealing with breaches of Exam Regulations: <https://www.nuigalway.ie/media/registry/exams/QA230---Procedures-for-Dealing-with-Breaches-of-Examination-Regulations.pdf>

Question 1:

- a) Explain the steps involved (in your own words) in a browser requesting and displaying a simple HTML webpage, located on a remote server.
(4 marks)
- b) Write HTML to recreate the following form (see Figure 1 below). The form is to enable users to submit a query to a website via its ticketing system. Each text box is spaced 5px apart from each other, please make sure when creating your HTML that you also maintain this spacing between input elements. The dropdown list contains fields for Technical Support, Sales, Accounts and Investors.

Submit your query and we'll be in contact as soon as possible



The form consists of four elements stacked vertically: an input field for 'Email Address', a dropdown menu currently showing 'Technical Support' with a downward arrow, a large text area for the query, and a button labeled 'Submit Ticket'.

Figure 1 Contact us form

- (7 marks)
- c) Do all HTML tags have a matching closing tag? Give a reason for your answer.
(4 marks)
- d) Create a simple menu for a local restaurant using HTML. The menu consists of separate headings for **starters**, **main courses** and **desserts**. An ordered list should be used to list 3 dishes under each heading. A final heading pertaining to **side dishes** should list three unordered side options the diner can choose from.

(5 marks)

20 Marks

PTO

Question 2:

- a) In the context of CSS style classes, explain in your own words what class and id-based selectors are. Give an example of each.
(5 marks)
- b) Using a CSS example describe how a single style can be applied to all tags of a particular type on a web page.
(5 marks)
- c) Explain in your own words how the grid system in Bootstrap works. In your answer discuss the purpose of rows and columns and describe how the grid facilitates responsive design.
(5 marks)
- d) Provide the HTML and corresponding Bootstrap classes to split a row into two separate sections, each sized 6 columns wide on both large and medium devices. In the case of smaller devices, the columns should be hidden.
(5 marks)

20 Marks

Question 3:

- a) Using JavaScript, write code to iterate through an array of objects (called myArray), removing objects placed at **odd** index positions in the array. Assume the array contains 8 vehicle objects which take the following form ({**make**: "Ford", **model**: "Focus", **age**:2}, ... , {**make**: "Toyota", **model**: "Yaris", **age**:5}). You may use the splice method to remove the objects from the array, however each time splice is called it will re-index the array. To counter this you should iterate through the array in reverse.
(9 marks)
- b) The splice method can be used to remove items from an array. Considering the above array which contains 8 vehicle objects, write code to show how the splice method can be used to delete the last three objects in the array.
(3 marks)
- c) Write JavaScript code that upon button click reads 3 numbers from 3 text boxes and doubles each number before saving it in the text box it came from.
(8 marks)

20 Marks

PTO

Question 4:

You have been asked to build a server backend using Firebase, which comprises a collection of RESTful APIs built using Firebase functions and a Firestore database. The backend is designed to support an Instagram style application. Each image shared by the users of the application should contain a description of the image, the upload time and date, the number of likes and the number of re-shares.

- a) Provide a sample JSON document which contains data for a single image shared on the platform.

(4 marks)

- b) Write the NodeJS code to create a server-side API using Firebase Functions which returns all image documents in the collection, sorted in **ascending** order by upload time.

(5 marks)

- c) Write the NodeJS code to create a server-side API using Firebase Functions which is used for accepting the image data sent via POST to the server. Assume that the request body contains JSON data. Prior to saving check the request body to see if there is a description. If no description is provided then the response to the user should be *image not saved* and no data should be saved in the database. If a description is provided then the data should be saved and the response to the user should be *image saved successfully*. Provide code to save the data in the request body to the Firestore database.

(6 marks)

- d) Write the NodeJS code to create a server-side API using Firebase functions which deletes individual documents from the collection by parsing the query string, extracting the id and removing the document found.

(5 marks)

20 Marks

END